

Lennard-Jones Potential

Conor Brolly
40400295

In this report we produce a simulation of atoms interacting through the Lennard-Jones pair potential. We compare estimates for the radial distribution function $g(r)$ for distances between pairs of atoms for two different temperatures.

1 Model

We investigate a system of interacting particles in a cube with periodic boundary conditions in every direction. Atoms are initialised in a face-centered cubic lattice with random velocities given by a Maxwell-Boltzmann distribution for given temperatures. The minimal unit cell is repeated in 3D $3 \times 3 \times 3$ times and the lattice constant is $2^{2/3}$.

The Lennard-Jones pair potential is given below and plotted in figure 1:

$$V(r) = 4 \left[\frac{1}{r^{12}} - \frac{1}{r^6} \right] \quad (1)$$

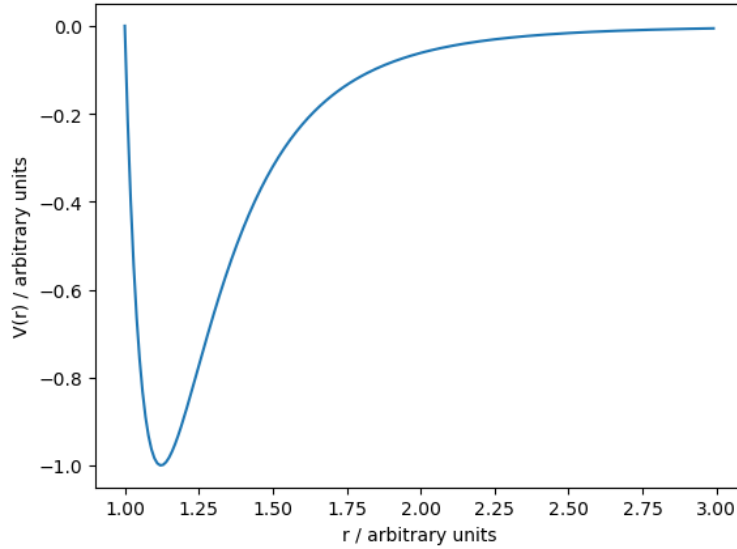


Figure 1: The Lennard-Jones pair potential as a function of distance between two atoms.

2 Monte-Carlo Simulation

To produce an estimate for the radial distribution function between atoms we run a molecular-dynamics simulation using the Python library ase (Atomic Simulation Environment). We perform a 10000 step Langevin dynamics simulation with a timestep of 0.005, temperature of $k_b T = 2.0$ and a friction coefficient of 1.0.

We can plot the energy of our system over the simulation to check for convergence, as given in figure 2:

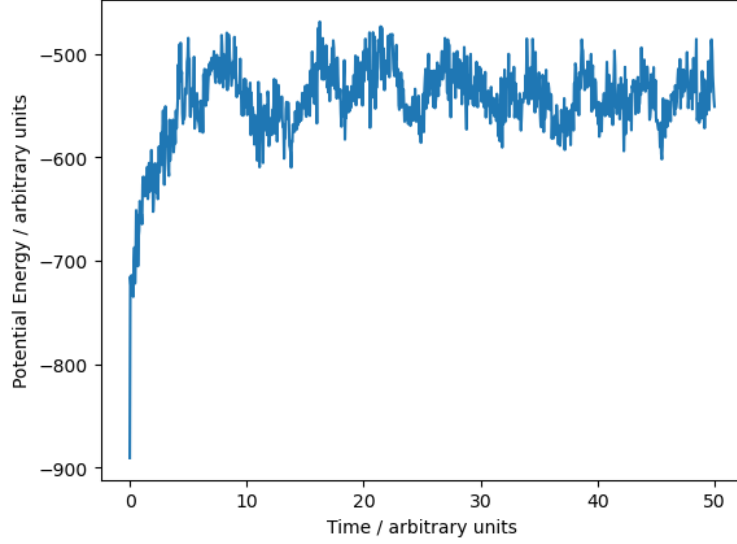


Figure 2: The energy of our system over 10000 iterations of a 0.005 timestep Langevin dynamics simulation at a temperature of $k_bT = 2.0$ and friction coefficient of 1.0. We see that the simulation reaches equilibrium at around 10 units of time.

Figure 2 shows that by 10 units of time the system reaches equilibrium. For any further calculations, we will discard this initial stage.

3 Radial Distribution Function

We can use our MD results to compute a histogram for the radial distribution - the distance between any two atoms. After normalising the results by correcting for the increasing volume at larger distances, we can produce an estimate for the radial distribution function between any two atoms.

Figure 3 shows us this radial-distribution function. There is a clear peak at around 1.1 units, with a secondary maxima at around 2.0 units.

We can also produce a radial distribution function estimate for a lower temperature of $k_bT = 0.5$ to compare the results. Figure 4 shows that, for lower temperatures, the minima and maxima in the distribution are more defined. This means that the atoms at lower temperatures are more structured, and that atoms become less structured as temperature increases.

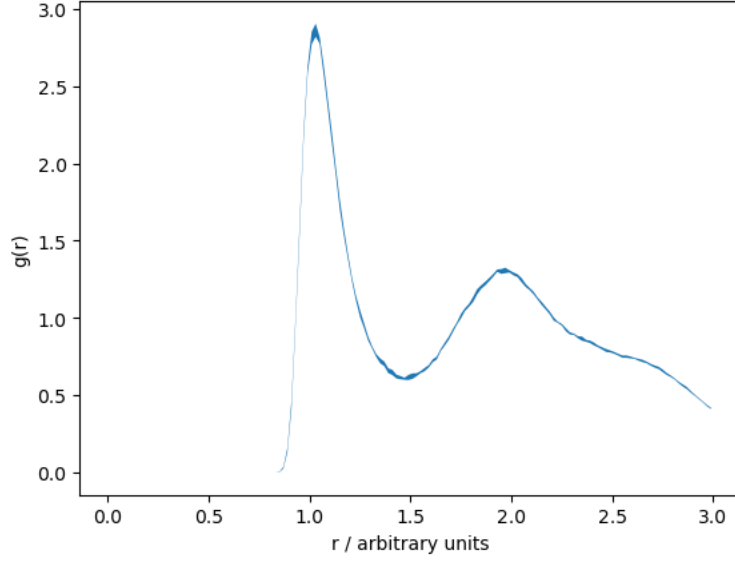


Figure 3: The estimated radial distribution function for distance between atoms in our system. The width of the lines represents the error corresponding to a 95% confidence interval on our estimate, which was estimated using block-averaging with a block-size of 100. The simulation was performed at a temperature of $k_bT = 2.0$.

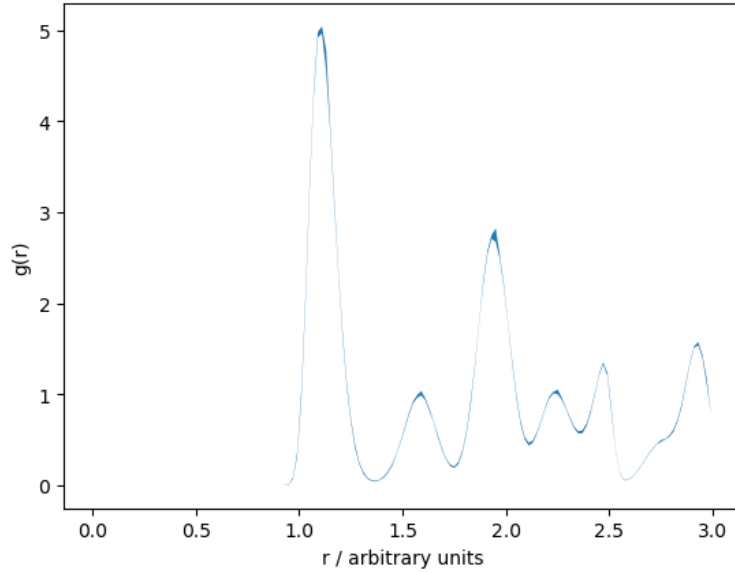


Figure 4: The estimated radial distribution function for distance between atoms in our system. The width of the lines represents the error corresponding to a 95% confidence interval on our estimate, which was estimated using block-averaging with a block-size of 100. The simulation was performed at a temperature of $k_bT = 0.5$.